

# STUDENT ENROLLMENT & COURSE MANAGEMENT SYSTEM

## 2. REQUIREMENT ANALYSIS

### 2.1 Project Overview

The Student Enrollment & Course Management System is a web-based application designed to simplify and automate student registration, course enrollment, and academic record management.

The system provides a centralized platform where students can register, log in, view available courses, enroll in courses, and check their enrollment status. Administrators can manage student information, add or update course details, monitor enrollment records, and generate reports.

### 2.2 Functional Requirements

| ID   | Requirement                   | Description                                                                                    |
|------|-------------------------------|------------------------------------------------------------------------------------------------|
| FR-1 | User Authentication           | Provide secure login and access to authorized students, faculty, and administrators.           |
| FR-2 | Student and Course Management | Create, maintain, update, and manage student records and course details.                       |
| FR-3 | Student Enrollment            | Allow students to enroll in available courses and maintain enrollment records.                 |
| FR-4 | Data Entry and Validation     | Capture required student and course information and validate entered values to prevent errors. |
| FR-5 | Enrollment Tracking           | Track student enrollment status and display relevant enrollment information.                   |
| FR-6 | Report Generation             | Generate reports related to student details, available courses, and enrollment information.    |
| FR-7 | Dashboard Visualization       | Display important student, course, and enrollment information through dashboards.              |
| FR-8 | User Role Management          | Control access according to user roles, permissions, and security configurations.              |

|       |                                |                                                                                                   |
|-------|--------------------------------|---------------------------------------------------------------------------------------------------|
| FR-9  | Course Availability Management | Display available courses and update course information when changes occur.                       |
| FR-10 | Record Management              | Store and retrieve student, course, and enrollment records efficiently in a centralized database. |

## 2.3 Non-Functional Requirements

| Category        | Requirement                                                                                                         |
|-----------------|---------------------------------------------------------------------------------------------------------------------|
| Usability       | Provide a simple and user-friendly interface for students, faculty, and administrators.                             |
| Security        | Use secure authentication, role-based access control, and permissions to protect student information.               |
| Reliability     | Maintain accurate student and course records and reduce the risk of data loss.                                      |
| Performance     | Target page loading, course searches, and report generation within approximately 3 seconds under normal conditions. |
| Availability    | Provide online access to authorized users whenever the system is available.                                         |
| Scalability     | Support growth in students, courses, enrollment records, and institutional users.                                   |
| Maintainability | Allow easy updates to course details, student records, and system configurations.                                   |
| Data Integrity  | Validate entered information and prevent duplicate or incorrect student and enrollment records.                     |

## 2.4 Data Flow Diagram (Logical Flow)

User → Student Enrollment & Course Management System UI → User Authentication → Student/Course/Enrollment Records → Data Validation → Database Relationships → Enrollment Processing → Reports → Dashboard.

## 2.5 Technology Stack

| Layer    | Technology            |
|----------|-----------------------|
| Frontend | HTML, CSS, JavaScript |

|                              |                                           |
|------------------------------|-------------------------------------------|
| Backend/Application Platform | Python Flask                              |
| Database                     | MySQL                                     |
| User Authentication          | Login and Role-Based Access Control       |
| Automation/Validation        | Form Validation and Backend Validation    |
| Analytics                    | Student Enrollment Reports and Dashboards |
| Development Tools            | Visual Studio Code                        |
| Web Browser                  | Google Chrome                             |

## 2.6 Main Data Entities

1. Student: Stores student ID, name, email, contact number, address, and other personal details.
2. Course: Stores course ID, course name, description, duration, fees, and course availability.
3. Enrollment: Maintains student enrollment details, course selection, enrollment date, and enrollment status.
4. Faculty: Stores faculty ID, name, email, contact details, and assigned courses.
5. Administrator: Manages student records, course details, enrollment information, and system operations.
6. User Account: Stores login credentials, user roles, and access permissions for students, faculty, and administrators.

## 2.7 Requirement Summary

- The system should provide secure login for students, faculty, and administrators.
- Students should be able to register and manage their profiles.
- Students should be able to view available courses and course details.
- Students should be able to enroll in their preferred courses.
- Administrators should be able to add, update, and manage student records.